CSC592: Generative AI Structured Literature Review (SLR)

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I. Introduction

The rapid advancement of Large Language Models (LLMs) has enabled artificial intelligences to perform tasks previously impossible for conventional machine learning systems. These models can interpret information, reason, and generate text at a scale and speed that far exceed human capabilities. However, as LLM capabilities continue to expand, so do concerns about their misuse. In particular, the generation of misinformation, false, misleading, or intentionally biased content, can now be automated, personalized, and spread at unprecedented levels.

While early research on the topic explored human-written misinformation and its societal and cognitive impacts, the emergence of AI-generated misinformation introduces new issues. LLMs can replicate stylistic features of journalism, commentary, and expert communication at a rapid pace, blurring the distinction between authentic and synthetic content. At the same time, current misinformation detection systems, typically trained for human-written misinformation, struggle to keep pace with rapid model development. As a result, researchers across multiple fields have begun investigating not only how AI-generated misinformation compares to human-written misinformation, but also how effectively current systems can detect it, and what datasets could be needed to study it systematically.

Despite growing interest in this topic, the literature remains fragmented. Few studies directly compare AI-generated and human-authored misinformation, while most work either focuses on detecting human-written misinformation or examines behavioral responses in isolation. Dataset papers create useful collections of AI-generated content, but they are rarely integrated into broader comparative or behavioral analyses. This fragmentation limits our understanding of how AI-generated misinformation functions, how users perceive it, and how effectively current detection systems respond to it.

A. Research Questions

This review is guided by the following research questions:

- **RQ1:** How does AI-generated misinformation differ from human-written misinformation in style, structure, and perceived credibility?
- **RQ2:** How effectively do current misinformation detection systems identify AI-generated misinformation?
- **RQ3:** How do users perceive and interact with AI-generated misinformation compared to human-written content?
- **RQ4:** What datasets exist for studying AI-generated misinformation, and what limitations do they exhibit?

II. Methodology

This Structured Literature Review (SLR) followed a transparent, multi-stage process based on PRISMA guidelines, including systematic searching, screening, eligibility evaluation, quality assessment, and thematic synthesis. The goal was to identify peer-reviewed research that examines the generation, characteristics, or perception of AI-generated misinformation within news or headline-based contexts.

A. Search Strategy

A comprehensive search was conducted across four major scholarly databases: IEEE Xplore, Scopus, PubMed, and the Consensus academic search engine. Searches were performed using Boolean combinations of keywords related to generative models and misinformation. Representative queries included: *("generative AI" OR "large language models") AND ("clickbait" OR "fearmongering" OR "misinformation")*, *"generative AI" AND "fake news"*, and *("generative AI" OR "large language models") AND ("clickbait" OR "fearmongering" OR "misinformation") AND "news headlines"*.

Across all sources, the search returned a total of 547 records. Only peer-reviewed publications from 2017 to 2025 were considered, ensuring relevance to the modern generative AI landscape.

B. Screening and Eligibility

Duplicate removal resulted in the elimination of 50 records, leaving 497 unique papers for title and abstract screening. These papers were evaluated against predefined inclusion and exclusion criteria. Inclusion criteria required that papers: (1) were published in or after 2017, (2) involved the generation of misinformation, (3) used or analyzed generative AI systems, and (4) focused on news or headline-based misinformation contexts.

Exclusion criteria removed papers that: (1) focused primarily on misinformation detection rather than generation, (2) were published before 2017, (3) examined fake-news classification systems without generative components, or (4) were not written in English. Based on these criteria, the title and abstract screening stage reduced the corpus from 497 records to 35 papers that were sufficiently relevant to warrant full-text examination.

During the full-text eligibility assessment, each of the 35 remaining papers was reviewed in detail to determine methodological completeness, relevance, and alignment with the review's objectives. Following this evaluation, 12 papers

met all eligibility requirements and were included in the final synthesis.

C. Quality Assessment

Each of the 12 included papers was evaluated using a structured rubric capturing four dimensions of research quality: (1) relevance to the research questions, (2) clarity of the problem definition, (3) methodological transparency, and (4) alignment between stated contributions and reported findings. This rubric ensured that included studies provided sufficient methodological detail to support synthesis and comparison across themes.

D. Study Characteristics: User Involvement

We also documented the level of user involvement across all 20 papers in the final corpus. As shown in Fig. 3, half of the studies (10 out of 20) included no users at all and focused solely on textual or model-centric analyses. Nine studies incorporated passive user involvement, typically through surveys or credibility ratings. Only one study employed active user participation. This distribution highlights a clear gap in the literature: despite the growing importance of user interaction in misinformation spread, very few studies assess how people actively engage with AI-generated misinformation in realistic contexts.

E. Data Extraction and Thematic Synthesis

The data extraction process focused on identifying the thematic relevance and conceptual contributions of each included study. Rather than extracting methodological or architectural details, each paper was examined to determine its alignment with the review's thematic categories and to document any gaps, limitations, or open problems highlighted by the authors. This approach emphasized the conceptual landscape of AI-generated misinformation research, allowing the review to capture areas of emphasis as well as recurring deficiencies across the literature.

Each paper was manually assigned to one of the predefined thematic categories based on its primary focus. The distribution of papers across these themes is shown in Fig. 2. These categorizations informed the structure of the subsequent analysis and helped identify recurring gaps in the literature.

The final thematic structure consisted of four categories:

- Human vs. AI Misinformation Comparison
- AI Misinformation Detection
- Human Perception of AI Misinformation
- Dataset Creation & Empirical Characterization of AI-Generated Misinformation

These themes form the structure of the subsequent analysis sections. The combined process provides a comprehensive and reproducible foundation for understanding how AI-generated misinformation is produced, how it differs from human-written misinformation, and how it is perceived and analyzed across disciplines.

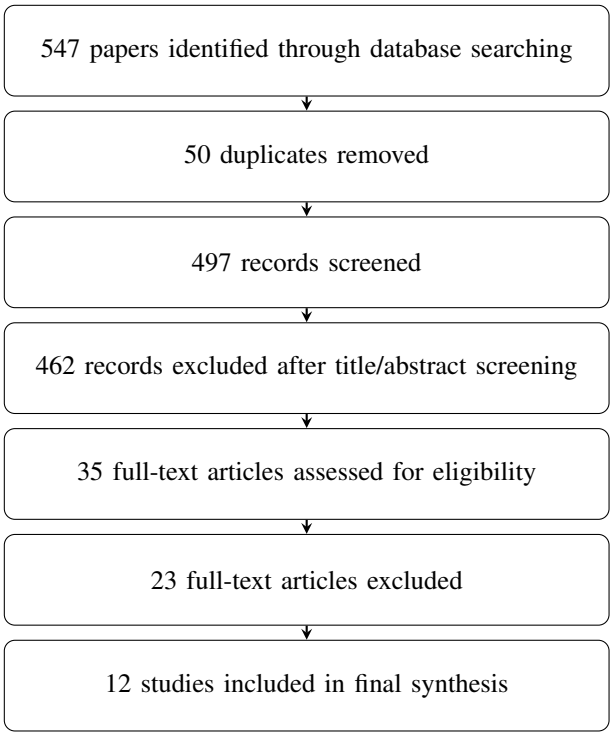

Fig. 1. PRISMA-style flow diagram of study selection process.

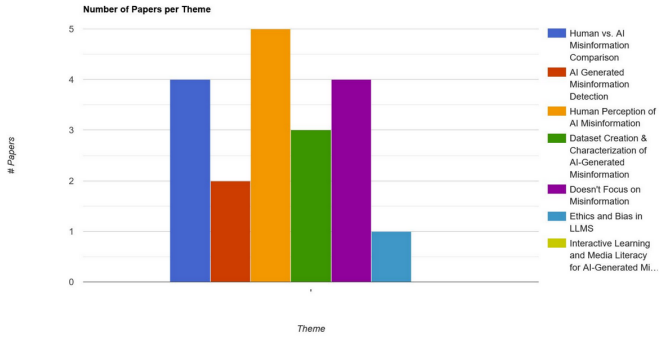

Fig. 2. Overview of the thematic categories identified in this Structured Literature Review.

III. AI vs. Human Misinformation

Several studies in the final selection [1], [2], [3], [4] directly compared the characteristics of AI-generated misinformation with human-authored misinformation. Across these works, a consistent pattern emerged: AI-generated misinformation often displays a higher level of linguistic polish, structural clarity, and rhetorical sophistication than its human counterparts. This distinction appears to stem from the inherent design of large language models, which prioritize coherence, completeness, and stylistic consistency in their outputs.

A recurring finding was that AI-generated misinformation often resembles professional journalism. Models produce more detailed and coherent narratives, frequently adding context, institutional references, or supporting evidence that human

misinformers do not typically include. These features give synthetic content a structured and authoritative appearance.

AI-generated misinformation also displayed a notable tendency to incorporate rhetorical markers of balance or uncertainty. For example, models often introduced disclaimers such as "the cause is not yet known" or "the individual has not been identified," elements that lend the appearance of journalistic caution and objectivity. This stylistic transparency, despite being applied to false or misleading content, can increase perceived trustworthiness by signaling neutrality. In addition, AI-generated stories frequently conclude with clear summaries or calls to action, creating more cohesive and persuasive narratives compared to the often fragmented nature of human-authored misinformation.

Another key difference observed is the use of evidence and sourcing. AI-generated misinformation consistently incorporated testimonial, statistical, or institutional references, including fabricated but realistic-sounding data or quotes from scientists, government organizations, or research institutions. While human misinformation often relies on emotional appeals, informal language, or anecdotal claims, AI-generated content more frequently mimicked the structure of evidence-based reporting. Some studies reported that AI-generated misinformation even exceeded the perceived credibility of human-written stories, with larger models producing outputs rated as equally credible or more credible than authentic news articles. These findings demonstrate that LLMs can generate misinformation that not only matches but sometimes surpasses human misinformers in perceived legitimacy.

Lastly, studies highlighted the scale and speed advantages inherent to AI-generated misinformation. LLMs can produce large volumes of coherent, tailored misinformation nearly instantaneously, enabling rapid dissemination and hypertargeting of persuasive narratives. Even interventions such as warning labels or disclaimers had limited effectiveness in reducing credibility judgments, and political partisanship remained a major driver of user perception regardless of content origin. Together, these findings show that AI-generated misinformation differs from human-authored misinformation not just in style and structure, but also in its potential for scale, perceived credibility, and persuasive impact.

IV. AI Misinformation Detection

Two studies in the final selection [2], [4] examined the challenges of detecting AI-generated misinformation, focusing on both automated systems and human-facing interventions. A central finding across these works is that current misinformation detection models, primarily trained on human-written misinformation, struggle to generalize to the linguistic and structural patterns produced by modern LLMs. Whereas most existing classifiers rely on stylistic cues, emotional valence, or overtly deceptive phrasing, AI-generated misinformation tends to avoid these signals, resulting in substantial drops in model recall despite maintaining high precision.

Studies found that automated detectors, trained on human-written misinformation, frequently miss AI-generated content due to its neutral tone, syntactic complexity, and blended factual elements. These characteristics bypass feature-based cues such as emotional language or overt deception. Detectors also over-weight institutional references, causing polished AI-generated content to appear trustworthy.

Another recurrent issue is the strategic use of authoritative references within AI-generated misinformation. Models frequently include mentions of government organizations, universities, health agencies, or scientific institutions. Zhou et al. found that approximately 77% of missed AI-generated misinformation cases included such references, suggesting that the appearance of legitimacy directly undermines classifier performance. This pattern highlights a critical weakness in both statistical and neural detection systems, which tend to over-weight perceived expertise cues and under-weight rhetorical structure when evaluating text credibility.

Human-facing interventions such as warning labels, community fact-checking cues, and literacy reminders were also tested with mixed results. While these interventions occasionally improved user skepticism, they did not reliably reduce the perceived credibility of AI-generated misinformation, particularly when the misinformation was framed in neutral or balanced language. Users continued to evaluate AI-generated content as equally or more credible than human-written misinformation, and political partisanship remained a major moderator of detection accuracy. These findings suggest that both automated and human-centric detection approaches face substantial challenges in keeping pace with rapidly improving generative models.

Overall, the literature indicates that current detection systems exhibit limited generalization to synthetic language patterns, struggle with the stylistic neutrality and high coherence of AI-generated misinformation, and are easily undermined by the inclusion of institutional references or mixed factual content. As generative models continue to improve in producing high-credibility narratives, detection gaps are likely to widen without more robust, adaptive, or provenance-based strategies.

V. Human Perception & Behavioral Responses

Several studies [5], [6], [7], [8] examined how users perceive and react to AI-generated misinformation. Across all works, participants struggled to accurately identify synthetic content, largely because AI-generated misinformation adopts linguistic cues associated with credible journalism. These texts frequently include institutional references, statistical claims, and balanced or hedged language, making them appear more factual and trustworthy than typical human-written misinformation.

Behavioral evidence shows that users are not only more trusting of AI-generated misinformation but also more likely to engage with it. In a dataset of over 91,000 posts, AI-generated misinformation received 8.19% more impressions, 20.54% more reposts, and 49.42% more likes than other misinformation types, indicating substantially higher levels of user interaction and virality compared to human-generated misinformation.

Intervention studies found that users had extremely low baseline accuracy in identifying misinformation (as low as 3.1%), but performance improved with warnings or explanations. Explanation-based interventions, especially GPT-4-generated explanations, consistently outperformed simple labels, improving accuracy by over 30% in general contexts and up to 97.6% in medical scenarios. However, some interventions increased both sharing and flagging behavior, suggesting mixed downstream effects. Overall, users perceive AI-generated misinformation as highly credible, interact with it at higher rates, and require structured, explanation-based support to reliably identify it.

VI. Dataset Creation & Empirical Characterization

Another study [9] focused on constructing and analyzing datasets that capture the distinctive properties of AI-generated misinformation. These works created collections of synthetic misinformation, mixed datasets combining AI- and human-written content, and large-scale social media datasets containing AI-generated posts. The goal across these efforts was to characterize linguistic and stylistic patterns, measure spread behavior, and provide resources for future detection and analysis tasks.

Despite these contributions, the literature highlights a significant shortage of publicly available datasets dedicated specifically to AI-generated misinformation. The few existing datasets differ substantially in domain and scope, ranging from news articles to social media posts and mixed-topic collections, which makes cross-study comparison difficult. Because the datasets cover different platforms, content types, and misinformation categories, they are not directly comparable or suitable as unified benchmarks. As a result, the field lacks standardized, comprehensive resources needed to track the evolution of AI-generated misinformation or to support the development of robust detection models.

VII. Gaps

Across the reviewed literature, several consistent gaps emerged. First, only a small number of studies directly compare AI-generated and human-written misinformation, leaving the field without a systematic framework for understanding how these two forms differ in credibility, structure, or impact. Research on detecting AI-generated misinformation is similarly limited and is often evaluated on narrow or domain-specific datasets, restricting generalizability as generative models evolve.

A more critical gap concerns user interaction. Nearly all existing studies rely on static surveys or passive credibility judgments, and only one study in the final corpus involved active user engagement with AI-generated misinformation. No work to date examines how users behave when they *actively generate, manipulate, or interact with* AI-produced misleading content in real time. This absence is striking given that modern generative systems enable interactive creation, rewriting, and personalization of misinformation on demand.

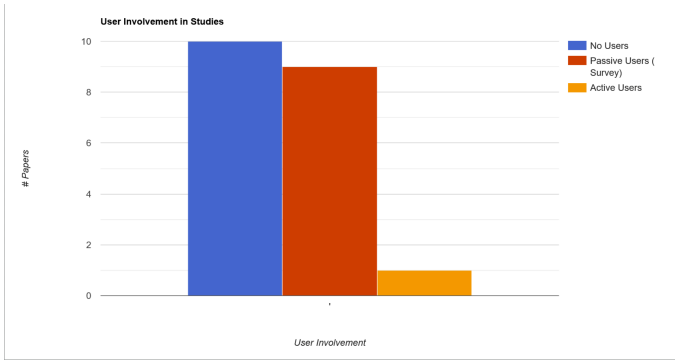

Fig. 3. Distribution of user involvement across 20 studies (some not included in final selection).

Furthermore, very few publicly available datasets exist for AI-generated misinformation, and existing collections span disparate domains such as news articles and social media posts, limiting reproducibility and cross-study comparison. Taken together, these gaps indicate that current research does not yet capture the dynamic, interactive nature of AI-driven misinformation ecosystems, creating a clear need for tools and studies that examine user behavior during live engagement with generative models.

VIII. Our Contribution: MisinfoLab

The findings of this review reveal a critical gap in current research: although modern generative models allow users to create misleading content interactively, virtually no studies examine how users behave and learn when actively generating or manipulating AI-generated misinformation. Existing work relies heavily on passive surveys, static headline evaluations, or credibility ratings, leaving unexamined the dynamic cognitive processes that occur during live engagement with generative systems.

To address this gap, this project introduces *MisInfoLab*, an interactive educational platform designed to help users understand and identify AI-generated misinformation through active creation, manipulation, and analysis. MisInfoLab translates the conceptual insights from this review, such as stylistic cues, rhetorical structures, and typical failure modes of detectors, into hands-on learning experiences. The platform is implemented as a Vite + TypeScript web application that integrates the Groq API for text generation and NewsAPI for real-world headline retrieval.

A. System Design

MisInfoLab consists of several integrated modules aligned with the thematic findings of this SLR (see Fig. 4). A *headline generator* allows users to input a headline or retrieve one from NewsAPI, after which they can choose a misinformation technique (clickbait, emotional triggering, political bias, or fabricated facts). The system then uses a large language model to rewrite the headline while highlighting the linguistic and rhetorical cues that make it misleading. A complementary *classifier module* attempts to identify the applied technique,

illustrating the challenges faced by automated detectors documented in prior literature.

All generative components in MisInfoLab use the `llama-3.1-8b-instant` model through the Groq API. This model was selected because it provides fast, coherent headline rewriting, question generation, and explanation generation, while producing polished outputs that closely resemble the stylistic patterns described in the literature on AI-generated misinformation. The system's detection component uses the `openai/gpt-oss-120b` model, which serves as a high-capacity classifier for identifying misinformation techniques. Using two different models—one for generation and one for detection—allows MisInfoLab to emulate real-world asymmetry between misinformation creation and defense mechanisms.

Additional features include a *side-by-side comparison view* of AI-generated versus human-authored headlines, a *story generator* that produces full articles derived from the misinformation headline, and a *learning page* where users can read about misinformation techniques. To support structured practice, a *quiz module* generates multiple-choice questions that test the user's ability to identify misinformation cues. Together, these components enable users to adopt both the role of misinformation creator and evaluator, offering a learning paradigm not present in current studies.

B. Technical Architecture

MisInfoLab is implemented as a client-side web application using Vite and TypeScript, with no backend server. All system functionality is executed directly in the browser, and all interactions with external models or data sources occur through client-initiated API calls. The architecture consists of three layers: the user interface layer, the API integration layer, and the local processing layer.

User Interface Layer: The interface is built entirely within the `src/` directory using TypeScript modules and standard HTML/CSS components. Each feature—headline generation, technique selection, side-by-side comparison, explanation view, story generation, and quiz evaluation—is implemented as a standalone UI component. The system does not employ a global state manager; instead, each component manages its own state through local variables that store the current headline, rewritten result, explanations, and quiz items.

API Integration Layer: All model inference is performed through direct browser calls to external APIs. The Groq API with the `llama-3.1-8b-instant` model is used for headline rewriting, explanation generation, and batch generation of ten quiz questions in a single request. The OpenAI `gpt-oss-120b` model is used exclusively for misinformation detection and receives only the manipulated headline for classification. Real-world headlines are retrieved from NewsAPI by performing a search for "America" restricted to articles published after October 2025. Because the system has no backend, all API responses are processed directly within the browser and rendered immediately in the UI.

Local Processing Layer: After receiving model outputs, the frontend performs lightweight post-processing, including JSON validation, formatting, and integration into the display components. All data for a user session—such as original headlines, rewritten versions, explanations, and quiz results—is stored in local component variables with no persistent storage or logging. The system follows a simple interaction flow: user action triggers an API request, the model response is returned to the client, and the interface updates accordingly. Ten quiz questions are generated in a single batch request, ensuring consistent difficulty levels and reducing API overhead.

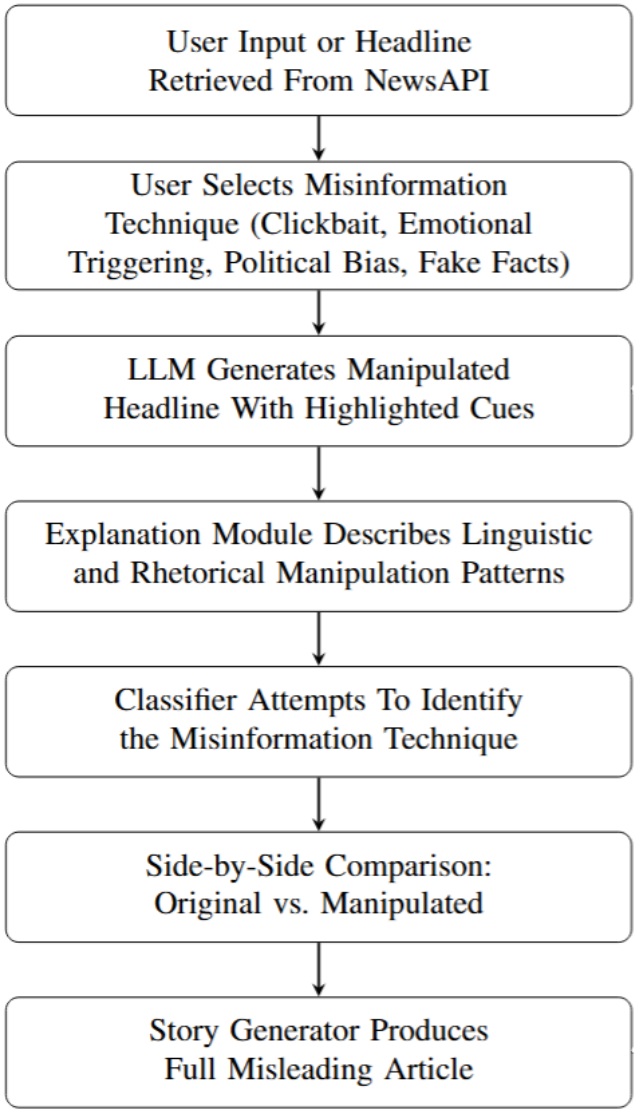

Fig. 4. System architecture diagram for MisInfoLab.

Overall, the architecture prioritizes simplicity, responsiveness, and full client-side execution. By eliminating backend complexity and performing all inference through external APIs, MisInfoLab remains lightweight while still enabling interactive experimentation with headline manipulation, explanation cues, and misinformation classification.

C. Evaluation Plan

As an educational platform, MisInfoLab is designed to improve users' ability to recognize misinformation techniques. Its effectiveness can be evaluated through a pre–post experimental design measuring accuracy gains in identifying clickbait, emotional manipulation, political bias, and fabricated facts. Users would complete an initial quiz, interact with the system's generation, comparison, and explanation modules, and then complete a post-quiz generated under similar difficulty conditions. Improvements in classification accuracy, reductions in misclassification rates, and increased confidence ratings can serve as primary metrics.

A secondary evaluation pathway involves behavioral logging: tracking which features users engage with (e.g., story generation, classifier inspection, cue highlighting) and correlating these interactions with learning outcomes. Although full deployment and data collection are beyond the scope of this project, this evaluation framework reflects how MisInfoLab could support future empirical studies on user behavior during real-time engagement with generative misinformation.

Overall, this project contributes both a structured review of the current research landscape and an interactive educational system that directly fills the largest identified gap: the absence of user-centered, creation-based learning tools for understanding AI-generated misinformation.

IX. Our Contribution: The Benchmark

In addition to the interactive platform, this project introduces a high-level *cross-model misinformation benchmark* designed to evaluate how different language models generate, interpret, and detect manipulative content. The benchmark uses a controlled set of neutral news headlines and a standardized prompting framework to elicit misinformation-style rewrites and obtain model-generated manipulativeness scores. Each model both produces rewritten headlines and evaluates the outputs generated by other models, creating a cross-comparison matrix that reveals patterns of agreement, divergence, and systematic bias in misinformation detection. This framework provides an initial step toward a reproducible, extensible methodology for studying how modern language models behave when confronted with subtle rhetorical manipulation. A dedicated section later in this paper presents initial empirical results and discusses their implications.

A. Benchmark Architecture

The benchmark is designed as a fully modular, model-agnostic pipeline. All models implement a lightweight interface consisting of two core methods: `generate_misinformation()` and `score_misinformation()`. This abstraction allows the system to treat every model uniformly, regardless of backend (cloud inference, local inference, quantized GGUF models, etc.). Prompts for generation and scoring are stored in an external JSON file, ensuring that all models receive identical instructions and enabling reproducible experimentation across runs.

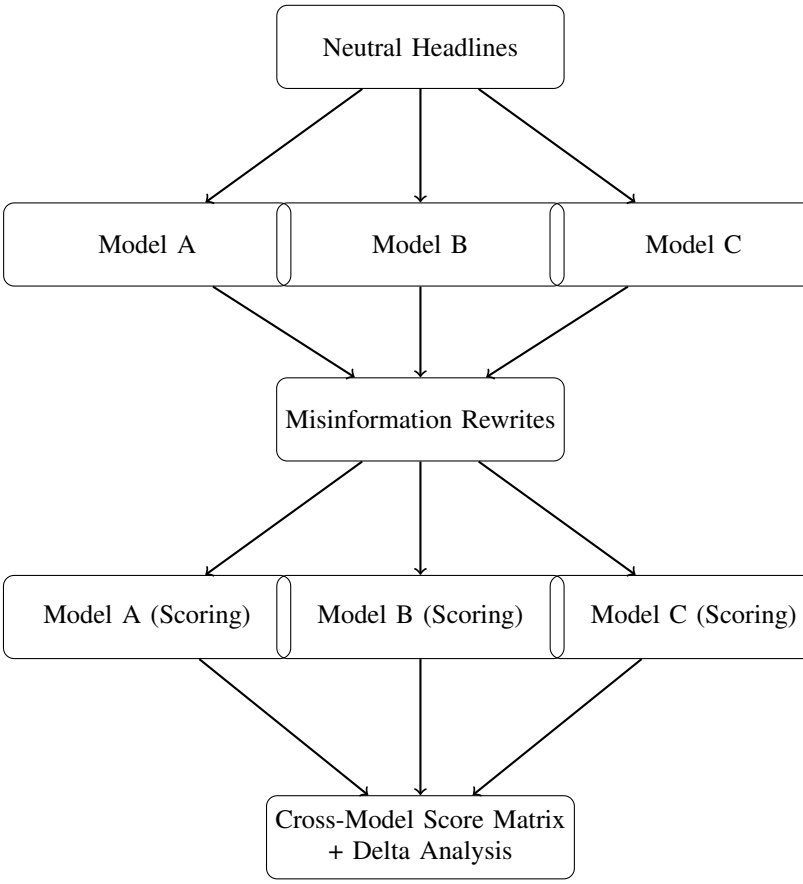

Fig. 5. Overview of the cross-model misinformation benchmark. Each model generates misinformation-style rewrites from neutral headlines, then all models score all outputs (including ground-truth headlines). The resulting cross-model matrix and delta analysis reveal systematic differences in sensitivity, bias, and calibration across models.

B. Plug-and-Play Model Integration

Adding a new model to the benchmark requires only (1) implementing the interface in a single Python class, and (2) registering that class in a configuration dictionary. No part of the benchmark logic needs to be modified. Once registered, the model is automatically included in all evaluation phases, generation, cross-model scoring, and delta analysis, making the system extensible with minimal developer effort.

C. Generation Phase

During the generation phase, each model rewrites a set of neutral news headlines into misinformation-style variants. The benchmark supports both full-scale runs and reduced subsets (the first 10 headlines) for fast debugging. Outputs are automatically cleaned, indexed, and stored in structured JSON files. This separation of generation from scoring enables repeatability: models can be re-scored without regenerating headlines, and new models can be added post hoc.

D. Cross-Model Scoring Phase

In the scoring phase, every model evaluates the manipulativeness of every other model's rewritten headlines. This

produces a dense cross-model evaluation matrix capturing how models perceive one another's outputs. The benchmark handles extraction of numerical scores, fallback parsing for unstructured model responses, and evaluation of ground truth neutral headlines to measure over-sensitivity.

E. Automation and Reproducibility

Running the benchmark is intentionally simple. Once models are registered, the entire evaluation can be executed via a small set of scripts: one for generation, one for scoring, and one for computing averages and producing visualization matrices. This design ensures that experiments are repeatable, scalable, and easy to extend, enabling researchers to rapidly compare models or integrate new ones as they become available.

X. Benchmark Results

A. Cross-Model Misinformation Scoring

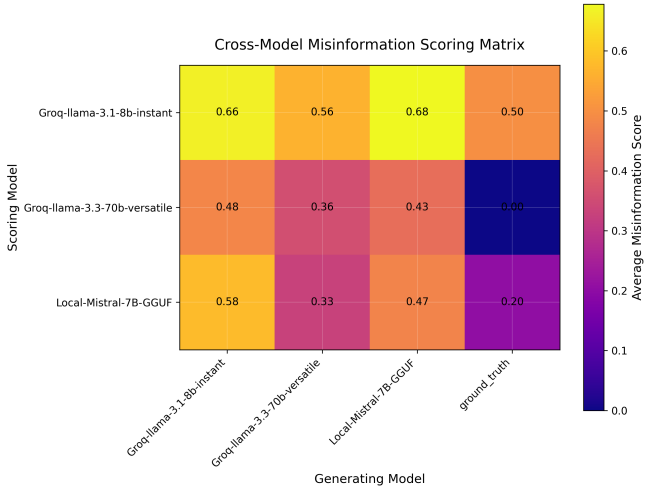

Fig. 6. Cross-model misinformation scoring matrix. Each cell represents the average manipulativeness score assigned by one model (row) to the headlines generated by another model (column). This visualization highlights systematic differences in model harshness, agreement patterns, and scoring bias.

Across the three scoring models, we observe notable differences in how manipulativeness is interpreted. The 8B model consistently assigns relatively high scores to all generated misinformation (0.56–0.68) and even rates the neutral ground-truth headlines at 0.50 on average—substantially higher than an ideal baseline—indicating a systematic oversensitivity to rhetorical cues. In contrast, the 70B model is the most conservative evaluator, producing much lower scores for generated content (0.36–0.48) and assigning a true baseline of 0.00 to the neutral headlines, suggesting strong calibration and a clear separation between neutral and manipulated text. The Mistral 7B model falls between these two extremes, with mid-range scores for generated examples (0.33–0.58) and a modest baseline of 0.21.

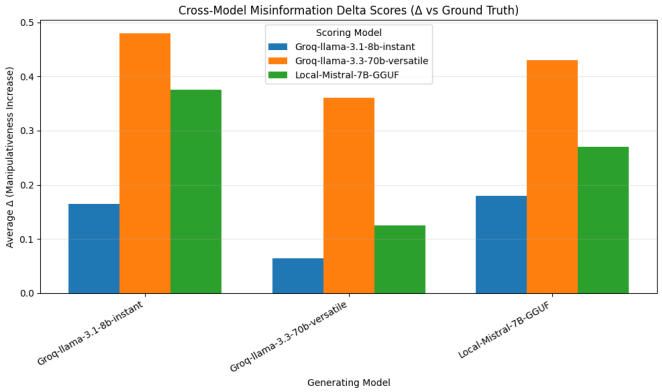

Fig. 7. Average deviation between each model's misinformation scores and the ground-truth baseline. Lower values indicate better discrimination between neutral and manipulated content, while higher values suggest oversensitivity or calibration errors.

B. Sensitivity to Neutral Headlines

The delta analysis, measuring how far each scoring model's evaluations deviate from the neutral baseline, highlights substantial differences across systems. The 8B model shows very small deltas when judging other models' outputs (0.06–0.18), suggesting that its inflated baseline of 0.50 compresses the scoring range and reduces its ability to distinguish subtle differences between generators. In contrast, the 70B model exhibits the largest deltas (0.36–0.48), reflecting its sharper discrimination between neutral and manipulated content; it consistently registers strong shifts away from the baseline when misinformation cues are present. Mistral 7B again occupies a middle ground (0.125–0.375), showing moderate sensitivity without the extremes seen in the other two models. Overall, these deltas illustrate that models with better-calibrated baselines (such as the 70B system) produce clearer separation between neutral and rewritten headlines.

XI. Conclusion

This review synthesizes the emerging literature on AI-generated misinformation, highlighting key differences between AI- and human-authored content, current challenges in automated detection, and the cognitive and behavioral factors that shape user susceptibility. Although research in this area is growing, the field remains fragmented, with limited comparative studies, few robust datasets, and notable gaps in user-focused educational approaches. These findings underscore the need for tools that both illuminate how AI-generated misinformation is produced and help users build the skills necessary to recognize it. MisInfoLab aims to address this need by translating the insights from this review into an interactive learning environment that supports misinformation literacy in the age of generative AI.

References

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APPENDIX A
STUDY SELECTION TABLE

TABLE I
STUDY SELECTION AND ELIGIBILITY ASSESSMENT (INCLUDED AND BORDERLINE STUDIES)

Authors	Year	Title	Venue	Included?	Reason / Notes
Kreps et al.	2022	All the News That's Fit to Fabricate: AI-Generated Text as a Tool of Media Misinformation	Journal of Experimental Political Science	Yes	Examines AI-generated misinformation compared to human-written misinformation
Zhou et al.	2023	Synthetic Lies: Understanding AI-Generated Misinformation and Evaluating Algorithmic and Human Solutions	ACM	Yes	Directly studies AI-generated misinformation and failures in detection
Danry et al.	2025	Deceptive Explanations by Large Language Models Lead People to Change Their Beliefs About Misinformation	ACM	Yes	Shows that deceptive LLM explanations increase belief in misinformation
Shin et al.	2025	Impact of Misinformation from Generative AI on User Information Processing	SagePub	Yes	Explores user cognition and how people process AI-generated misinformation
Drolsbach & Prollochs	2025	Characterizing AI-Generated Misinformation on Social Media	arXiv	Yes	Analyzes stylistic and engagement differences between AI- and human-generated misinformation
Danry et al.	2024	Deceptive AI systems that give explanations are more convincing than honest AI systems and can amplify belief in misinformation	arXiv	Yes	Shows deception amplification effects from LLM-generated misinformation explanations
Lucas et al.	2024	The Longtail Impact of Generative AI on Disinformation: Harmonizing Dichotomous Perspectives	IEEE	Yes	Examines large-scale disinformation propagation enabled by generative AI
Shah et al.	2024	Navigating the Web of Disinformation and Misinformation: Large Language Models as Double-Edged Swords	IEEE	Yes	Covers both generating and detecting misinformation using LLMs
Dash et al.	2025	The Persuasive Potential of AI-Paraphrased Information at Scale	Oxford Academic	Yes	Shows how generative AI paraphrasing increases misinformation persuasiveness
Pillai et al.	2025	Engagement-driven Persona Prompting for Rewriting News Tweets	ACL	Yes	Uses LLMs to rewrite headlines/tweets to maximize emotional response; relevant to manipulative content
McLoughlin	2023	A Framework for Understanding the Weaponisation of the Internet	IEEE	Maybe	Broadly discusses misinformation/disinformation spread; includes generative AI as a mechanism
Momeni	2025	Artificial Intelligence and Political Deepfakes: Shaping Citizen Perceptions Through Misinformation	SagePub	Maybe	Focuses on AI-generated political deepfake misinformation; relevant but not text-based

APPENDIX B
PROMPTS USED FOR GENERATION AND DETECTION

A. Headline Generation and Explanation Prompt

```
You are an AI that rewrites headlines to demonstrate misinformation techniques.

Selected Technique: ${mode}

GENERAL RULES (always apply):
1. Make the rewritten headline punchy, emotionally manipulative, and misleading.
2. Bold ONLY the manipulative/emotional words using **markdown bold**.
   - Do NOT bold the entire headline.
   - Do NOT wrap the whole sentence in asterisks.
   - Only bold the misleading/emotional words.
3. Return JSON only.

SPECIAL RULES FOR POLITICAL BIAS:
If mode == "Political Bias", then:
- Choose a clear partisan framing (LEFT-leaning or RIGHT-leaning).
  Randomly pick one per generation.
- Use emotionally loaded, ideological language.
- Blame or praise a political group, party, ideology, or public figure.
- Insert partisan narratives (e.g., corruption, incompetence, elitism, extremism).
- Slightly distort context to favor one side.
- Ensure **at least 3** manipulative/emotional terms are bolded.
- Avoid neutrality  pick a side clearly.

SPECIAL RULES FOR FAKE FACTS:
If mode == "Fake Facts", fabricate authoritative-sounding details that are false.

SPECIAL RULES FOR EMOTIONAL TRIGGERING:
If mode == "Emotional Triggering", emphasize fear, anger, outrage, or panic.

SPECIAL RULES FOR CLICKBAIT:
If mode == "Clickbait", exaggerate stakes and add urgent/dangerous language.

IMPORTANT STYLE CORRECTION:
- If mode is NOT "Clickbait", avoid using clickbait-style phrases:
  "shocking", "devastating", "explosive", "you wont believe",
  "stuns", "panic", "outrage", "urgent warning".
- Keep manipulation focused on the selected technique.
- For Political Bias: use partisan framing, not dramatic stakes.
- For Fake Facts: invent details, not danger.
- For Emotional Triggering: target a specific emotion.
- For Out-of-Context Distortion: change context subtly.

JSON schema:

{
  "headline": "REWRITTEN HEADLINE WITH **BOLDED KEYWORDS ONLY**",
  "explanation": "SHORT EXPLANATION OF WHY THIS REPRESENTS '${mode}' MISINFORMATION"
}

Original Headline:
"${text}"
```

B. Detection and Scoring Prompt

```
You are an AI trained to evaluate multiple specific misinformation techniques.

Important:
- Do NOT label emotionally intense headlines as "clickbait" unless the headline
  uses true clickbait-style exaggeration.
- Evaluate each technique independently.

DEFINITIONS:
Clickbait = sensationalism used to generate curiosity or shock
Political Bias = language favoring or attacking a political group
Emotional Triggering = fear/anger/sadness language WITHOUT exaggerated stakes
Fake Facts = fabricated or impossible details made to deceive
Out-of-Context Manipulation = true information framed misleadingly

Return ONLY JSON:

{
  "misinformation_score": number 0-100,
  "emotional_triggering": number 0-100,
  "clickbait": number 0-100,
  "political_bias": number 0-100,
  "fake_facts": number 0-100,
  "out_of_context": number 0-100,
  "manipulative_phrases": ["..."],
  "psychological_triggers": ["..."],
  "reasoning": "..."
}

Headline:
"${headline}"
```